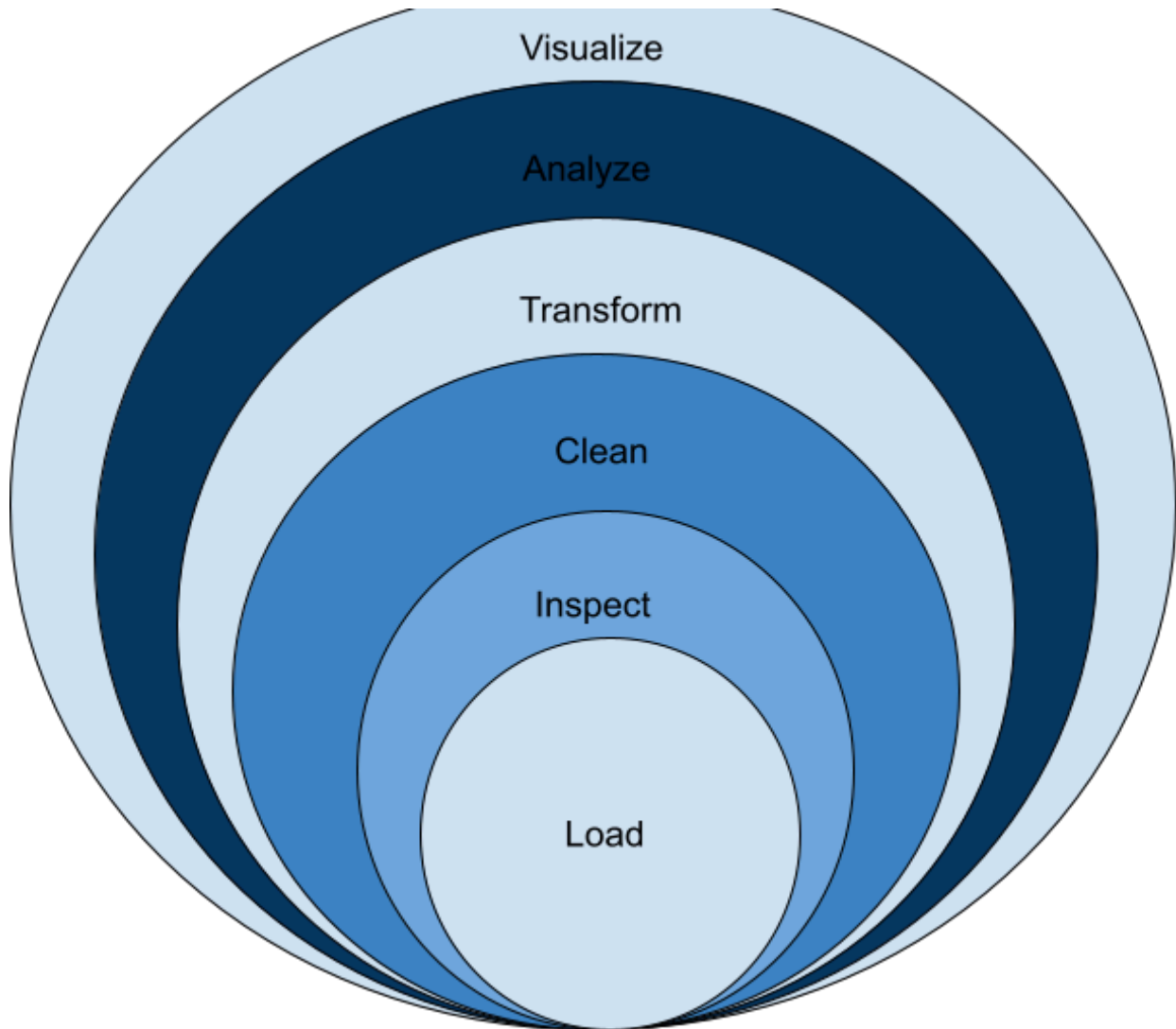


Python for data analysis dataset - Weather History

Tool: Google Colab (Python)

Digramatic sketch:-



Step 1: Load the data

This is the starting point of any data analysis as we need to bring the data into python before we do anything with it. Thus I loaded the **weather csv file into python using pd**. The dataset had **96,453 rows and 12 columns covering 10 years** of hourly weather recordings.

Step 2: Checked for missing values

This is the 2nd step, where missing data can break any model we build later on. Thus I checked every column for empty or missing values using **df.isnull.sum()** . Only one column had missing data where the precip type had 517 empty cells.

Step 3: cleaned the data

This is the 3rd step, where we clean the dataset to obtain a correct prediction . Thus I filled the **517 missing precip type values with the word "unknown"** instead of deleting those rows and also **found & removed the 24 duplicate rows** that were exact copies of the other rows using **df.drop_duplicates()**

Step 4 : Convert the data type

This is the 4th step, where we extract useful information like month & year of each recording. Thus the data column was stored as plain text so python was unable to understand if it was a real date or just a text. So I converted that to a proper date format using **pd.to_datetime**.

Step 5 : Create a new calculated column

This is 5th step where we bring the total count using **df['Temp (F)'] = (df['Temp (C)] * 9/5) + 32**. Thus I created a new column called Temp (F) by converting the existing celsius values to Fahrenheit using the above formula. The total column count was now 12 to 15.

Step 6 : Group by analysis

This is the 6th step where we create a group by analysis using **df.groupby()**. Thus I grouped the data by month & calculated the avg temp for each month. This turned the 96,000 rows data into 12 simple numbers. The final result was a clear seasonal pattern showing Jan avg 0.82 degree celsius and July avg 22.97 degree celsius confirming a 4 season climate.

Step 7 : Filtering

This is the 7th step where we filter only the important one which are useful within a large dataset for specific conditions. Thus I filtered the dataset to find all recordings where wind speed exceeded 30 km/h. This returned me **1,562 records** of storm events.

Step 8 : Sorting

This is the 8th step where we sort the data according to our conveniences. Thus I sorted the dataset to find the top 5 hottest days ever recorded on the scale. The **hottest** was **22nd july 2007 at 39.91 degree celsius** with very low humidity of just 0.13.

Step 9 : Correlation

This is the final step where we correlate each column of data to make a relationship between. Thus I calculated the corr between the temp and all other numerical variables ; which resulted in confirming that **humidity has the strongest relationship with temp at -0.63**, meaning the hotter days are consistently direr. Here the wind speed showed me almost no relationship at 0.01.

Step 10 : Charts display

